Deepfake Audio Detection

Real vs. Synthetically Cloned Voices

AI-600 Deep Learning · Midterm Report · Spring 2026

Submitted: 08 May 2026

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Abstract

The proliferation of neural voice cloning systems has made it trivially easy to synthesize speech perceptually indistinguishable from a target speaker, enabling large-scale voice fraud, political misinformation, and authentication bypass at scale. Recent evaluations confirm that LLM-based text-to-speech architectures now produce synthetic audio that defeats systems effective against conventional vocoders, underscoring the urgency of this problem. This project develops a binary classifier to distinguish genuine human speech from synthetically generated audio using the ASVspoof 2019 Logical Access (LA) benchmark — a dataset of 121,000+ utterances spanning 19 TTS and voice conversion attack types. Our approach builds on end-to-end architectures (RawNet2, AASIST) and self-supervised feature representations from wav2vec 2.0, augmented with RawBoost data augmentation to improve generalization to unseen spoofing attacks — the core challenge identified across the benchmark literature. As of the midterm submission, the full dataset has been acquired and verified through comprehensive exploratory data analysis. RawNet2 baseline implementation is underway, with full AASIST integration planned for the remainder of the project.

1. Introduction

The deepfake audio threat has evolved from a theoretical concern to a practical attack vector. Singh et al. [16] evaluate detection systems against LLM-based TTS architectures — including Dia2 and Maya1 — and find that models effective against conventional vocoder-based synthesis fail significantly when confronted with newer LLM-based generation methods. Voice-cloning services now operate commercially at low cost, enabling impersonation attacks on voice-authenticated banking systems, injection of synthetic speech into judicial proceedings, and automated generation of political misinformation at scale. This threat landscape creates an urgent demand for robust, generalizable countermeasures that extend beyond controlled laboratory conditions.

The ASVspoof challenge series [2,3] has driven systematic benchmarking of spoofing countermeasures since 2015, with the 2019 edition introducing 19 distinct TTS and voice conversion attack types across the Logical Access (LA) track. The LA benchmark is deliberately structured to stress-test generalization: the training and development partitions expose only six known attacks (A01–A06), while the evaluation partition introduces 13 previously unseen attacks (A07–A19) [2]. Müller et al. [10] formalize the severity of this challenge: models achieving sub-5% Equal Error Rate (EER) on the ASVspoof 2019 LA evaluation set suffer performance degradation of up to 1,000% when evaluated on real-world deepfake recordings of public figures, demonstrating that benchmark performance and real-world generalization remain fundamentally decoupled.

This project directly addresses the generalization challenge by implementing and evaluating a progression of detection architectures on the ASVspoof 2019 LA benchmark. Beginning with the RawNet2 end-to-end baseline [4], the project advances to the AASIST graph-attention architecture [5] augmented with self-supervised feature representations from wav2vec 2.0 [7], following evidence that fine-tuned SSL front-ends substantially improve cross-condition robustness [8]. RawBoost data augmentation [9] is applied to simulate telephony channel effects and real-world noise corruption. System performance is measured using EER and tandem Detection Cost Function (t-DCF) [3] on the LA evaluation partition. Section 2 reviews the relevant literature. Section 3 describes the proposed methodology. Section 4 details the experimental setup. Section 5 presents current progress and preliminary findings. Section 6 outlines the remaining plan, and Section 7 concludes.

2. Literature Review

2.1 Datasets and Evaluation Benchmarks

The ASVspoof challenge has served as the primary driver of anti-spoofing research since its inception in 2015, with successive editions escalating the diversity and realism of synthetic attacks. The ASVspoof 2019 LA dataset [2] contains 121,000+ utterances from 107 speakers drawn from the VCTK corpus, with spoofed speech generated by 19 TTS and voice conversion systems (A01–A19). The dataset employs a speaker-disjoint design: no speaker appears in more than one partition, ensuring that models cannot exploit speaker identity as a shortcut to correct classification.

The ASVspoof 2019 challenge [3] established two official evaluation metrics: the Equal Error Rate (EER) — the operating threshold θ^* at which the false alarm rate equals the false rejection rate ($FAR(\theta^*) = FRR(\theta^*)$) — and the tandem Detection Cost Function (t-DCF), which measures countermeasure cost within a complete speaker verification pipeline. Both metrics are lower-is-better, and evaluation is always reported on the evaluation partition. The official baselines are CQCC+GMM (EER: 9.57%, min t-DCF: 0.2366) and LFCC+GMM (EER: 8.09%, min t-DCF: 0.2116) [3], which 27 of 48 participating teams were able to surpass.

2.2 Feature Extraction Approaches

Early spoofing countermeasures relied on handcrafted spectral features — MFCC, LFCC, and CQCC — which capture vocoder artefacts through frequency domain analysis. While computationally efficient, these representations generalize poorly: LFCC-based systems collapse catastrophically on out-of-domain recordings, degrading from 2.98% to 29.42% EER between ASVspoof 2019 and 2015 evaluation conditions [8]. A sub-band masking analysis reveals that LFCC relies on attack-specific high-frequency cues (5.6–8 kHz), whereas self-supervised learning (SSL) features leverage the linguistically meaningful 0.1–2.4 kHz band — explaining the cross-condition advantage of SSL representations.

Baevski et al. [7] introduce wav2vec 2.0, a framework that masks speech in the latent space of a 7-block CNN encoder and learns contextualized representations through contrastive learning over quantized latents across a 12- or 24-block Transformer. Pre-trained on up to 60,000 hours of unlabelled speech, it produces layered representations where shallow layers retain acoustic detail and deeper layers encode phonetic and semantic content. Wang and Yamagishi [8] establish that when a fine-tuned wav2vec 2.0 XLSR front-end is combined with a deep LSTM-based backend, the system achieves 0.11% EER on ASVspoof 2019 LA. Crucially, fine-tuning end-to-end is essential: frozen representations yield 1.47% EER under the same conditions.

2.3 Detection Architectures

The detection architecture literature has evolved from CNN-based spectrogram classifiers toward end-to-end and graph-structured models. Tak et al. [4] introduced the first end-to-end anti-spoofing system (RawNet2) that processes raw waveforms directly through a bank of fixed sinc-parameterized filters — deliberately non-learnable to avoid overfitting to the six known training attacks — followed by six residual blocks with Feature Map Scaling (FMS) attention and a GRU layer for temporal modelling. The RawNet2 single system achieves 4.66% EER on the ASVspoof 2019 LA evaluation set; score-level fusion with an LFCC-GMM baseline improves this to 1.12% EER.

AASIST [5] advances the architecture to a heterogeneous graph neural network. A RawNet2-style encoder feeds spectral and temporal artefact representations into separate graph branches processed by stacked Heterogeneous-Stacking Graph Attention Layers (HS-GALs). A max graph operation (MGO) introduces competitive node selection, and a stack node aggregates cross-domain artefacts before a fully-connected scoring head. Even the lightweight AASIST-L variant, with only 85K parameters, achieves 0.99% EER — outperforming all competing single systems at the time of publication [5]. The 2026 resolution-aware detection framework [15] achieves 0.16% EER through multi-resolution spectral alignment with cross-scale attention, establishing the current performance frontier.

2.4 Generalization and Robustness

The dominant finding of the deepfake audio detection literature is a profound generalization gap between controlled benchmark performance and real-world detection capability. Müller et al. [10] evaluate 12 architectures on ASVspoof 2019 and a 37.9-hour In-the-Wild (ITW) dataset of celebrity deepfakes sourced from social media. The best-performing ASVspoof system (RawGAT-ST, 1.23% EER) degrades to 37.15% EER on ITW data — a degradation of over 1,000%. Critically, adding ITW data to training does not substantially improve ITW performance (33.94% $\rightarrow$ 33.14% EER), indicating that the problem is distributional mismatch rather than data quantity.

Recent 2025–2026 work has begun to systematically address this gap. A generalizable artifact-focused approach [11] proposes explicit Artifact Detection Modules (ADMs) that disentangle synthesis artifacts from speaker identity, achieving an F1 score of 0.813 on the ITW dataset. Nguyen-Le et al. [18] introduce AFSS (Artifact-Focused Self-Synthesis), which generates pseudo-fake samples at training time to enforce artifact-centric learning, achieving a mean EER of 5.45% across seven evaluation datasets — the strongest published generalization result as of early 2026.

2.5 Loss Functions and Reproducibility

Wang and Yamagishi [6] train each configuration of front-end, back-end, and loss function six times with different random seeds and apply Holm–Bonferroni-corrected paired z-tests. They show that single-seed comparisons are statistically unreliable: the same architecture spans 1.92%–3.10% EER across seeds. Their proposed P2SGrad loss — mean-squared error on cosine similarities with zero hyperparameters — consistently outperforms tuned AM-Softmax and OC-Softmax. Li et al. [1] corroborate that OC-Softmax and P2SGrad outperform standard cross-entropy on cross-dataset evaluation, and identify knowledge distillation as the most effective technique for closing the in-domain/out-of-domain gap.

2.6 Gap Analysis

Three observations motivate the approach taken in this project. First, the strongest single-system result on ASVspoof 2019 LA (AASIST, 0.83% EER) does not translate to real-world conditions: Müller et al. [10] report 37%

EER for the same architecture on in-the-wild deepfakes. Second, while fine-tuned wav2vec 2.0 systems achieve the lowest EER, they require 24–40 GB VRAM that is impractical for a constrained student project, and their min t-DCF can paradoxically be worse than the LFCC baseline despite lower EER [8]. Third, AASIST-L offers a compelling middle ground: SOTA-competitive accuracy at 85K parameters with direct compatibility with on-the-fly RawBoost augmentation. These findings directly motivate the proposed pipeline.

3. Proposed Methodology

3.1 Problem Formulation

Deepfake audio detection is formulated as a binary classification problem. Given an input waveform **x** of variable length, a detection function f: **x** → s maps the input to a real-valued detection score s, where lower scores indicate genuine (bonafide) speech and higher scores indicate synthetic (spoofed) speech. A decision threshold θ determines the binary output:

Decision Rule

```
■ = 0 (bonafide) if s < θ
■ = 1 (spoof) if s ≥ θ
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EER threshold θ chosen where FAR(θ*) = FRR(θ*)*

3.2 Feature Extraction

The methodology investigates two complementary feature extraction pipelines operating in parallel:

- **End-to-End Stream.** Raw waveforms at 16 kHz are processed directly by a bank of sinc-parameterized filters, following the RawNet2 design [4]. These filters are initialized with Mel-scale frequency spacing and kept fixed (non-learnable) to prevent overfitting to the frequency characteristics of the six known training attacks — a design choice explicitly motivated by generalization.
- **SSL Stream.** Raw waveforms are additionally processed through a pre-trained wav2vec 2.0 encoder [7]. Features are extracted from intermediate transformer layers rather than the final layer, as intermediate representations retain more acoustically grounded anti-spoofing information [13]. The SSL encoder is fine-tuned end-to-end with the detection backend; frozen representations serve only as an ablation baseline [8].

All input waveforms are processed as 4-second (64,000-sample at 16 kHz) segments. Variable-length utterances are padded with silence or randomly cropped to this fixed length during training. The unified pipeline is summarized below:

```
Raw Waveform (16 kHz) | |---> Sinc Filters (fixed) --> 6 Residual Blocks (FMS) --> GRU(1024) |
| '---> wav2vec 2.0 (intermediate layers, fine-tuned) | | | Heterogeneous Graph (AASIST
HS-GAL) | | | Max Graph Pool --> Readout <--------------------------- | Detection Score (s)
```

3.3 Model Architecture

Baseline — RawNet2 [4]. RawNet2 processes raw waveforms through a bank of sinc filters parameterized by fixed lower and upper cutoff frequencies arranged on the Mel scale. Six residual blocks equipped with Feature Map Scaling (FMS) attention follow, where FMS applies filter-wise scaling to emphasize informative frequency bands. A GRU layer with 1,024 units captures temporal dependencies before a fully-connected output layer produces a two-class detection score. The full system achieves 4.66% EER on ASVspoof 2019 LA as a single system.

Proposed — AASIST [5]. AASIST extends the RawNet2 encoder with a heterogeneous graph neural network designed to jointly model spectral and temporal artefacts, layer-by-layer:

- **Sinc-conv front-end:** 70 fixed Mel-scaled sinc filters (kernel length 129), followed by max-pooling, batch normalization, and SeLU activation.
- **Residual encoder:** Six residual convolutional blocks (first two with 32 filters, last four with 64 filters), each followed by max-pooling and batch normalization. Output is a feature map $F \in R^{C \times S \times T}$ representing channels, spectral bins, and temporal frames.
- **Graph construction:** Two graphs are derived from F. The spectral graph G_s uses temporal-max-pooled node features; the temporal graph G_t uses spectral-max-pooled node features.
- **HS-GAL layers:** Three distinct projection vectors compute attention weights for intra-spectral, intra-temporal, and cross-domain edges. A learned stack node aggregates cross-domain context analogously to a [CLS] token.
- **Max Graph Operation (MGO):** Two parallel HS-GAL branches each apply two HS-GAL layers and graph pooling (50% spectral and 30% temporal node removal), with element-wise maximum taken across branches.
- **Readout:** Final utterance representation is the concatenation of node-wise max-pooling, average-pooling, and the stack node embedding, projected to 2 logits.

The full AASIST-L model contains approximately 85,000 trainable parameters and fits within 8 GB of GPU memory at the chosen batch size.

3.4 Data Augmentation — RawBoost

To improve robustness to channel effects and real-world acoustic degradation, RawBoost [9] augmentation is applied during training. RawBoost operates directly on raw waveforms without requiring external noise recordings, using three stochastic noise components:

- ① **Linear and non-linear convolutive noise (LnL-CM):** Models transmission channel effects using Hammerstein systems with five notch filters spanning 20–8,000 Hz.
- ② **Impulsive signal-dependent additive noise (ISD-ADD):** Simulates burst noise with a relative power ratio in the range 0–10% of samples per utterance.
- ③ **Stationary signal-independent additive noise (SSI-ADD):** Adds FIR-coloured Gaussian noise at 10–40 dB SNR.

Tak et al. [9] demonstrate that combining modes ①+② achieves a 44% reduction in EER and a 27% relative reduction in min t-DCF on ASVspoof 2021 LA, outperforming SpecAugment, WavAugment, and codec augmentation. Mode ③ is retained as an ablation condition. Augmentation parameters are re-sampled per-utterance per-epoch so that the model never sees the same distorted waveform twice.

3.5 Training Configuration

Models are trained using weighted binary cross-entropy to address the ~9:1 class imbalance. The Adam optimizer is used with an initial learning rate of 1×10^{-4}, with cosine annealing over 100 epochs for AASIST following the original training protocol [5]. Mini-batches of 32 utterances are used, each waveform randomly cropped to 64,000 samples. Training runs for up to 100 epochs with early stopping on development EER (patience: 5 epochs). All experiments are repeated with three random seeds, with results reported as mean ± standard deviation, following the multi-seed protocol of Wang and Yamagishi [6].

3.6 Evaluation Metrics

Primary evaluation uses the official ASVspoof 2019 LA scoring toolkit. Two metrics are reported on the evaluation partition:

- **Equal Error Rate (EER, %):** The operating point at which the false acceptance rate and false rejection rate are equal.
- **Minimum tandem Detection Cost Function (min t-DCF):**

$$\text{min t-DCF}_{norm} = \min_s \left[\beta \cdot P^{cm}_{miss}(s) + P^{cm}_{fa}(s) \right]$$

where β is set per-attack from the ASV system's false acceptance behaviour, and P^{cm}_{miss}, P^{cm}_{fa} are the miss and false-alarm rates of the countermeasure at threshold s. Both metrics are reported for the pooled evaluation set and per-attack.

4. Experimental Setup

4.1 Dataset

All experiments use the ASVspoof 2019 Logical Access (LA) track [2,3]. The dataset is downloaded from the Edinburgh DataShare repository (doi:10.7488/ds/2461) and organized into three speaker-disjoint partitions. Our exploratory data analysis confirmed the following statistics from the protocol files:

Split	Bonafide	Spoofed	Total	Speakers	Attacks
Train	2,580	22,800	25,380	20	A01–A06
Development	2,548	22,296	24,844	20	A01–A06
Evaluation	7,355	63,882	71,237	67	A07–A19

Table 1: ASVspoof 2019 LA dataset split statistics.

Speaker overlap analysis confirmed zero intersection between all three partitions. The class imbalance is consistent across splits, with spoofed utterances comprising approximately 89.8% of each partition — motivating the use of weighted loss functions. Attack A17 (VAE-based voice conversion) and A10 (Tacotron2 + WaveRNN) have been identified as the hardest evaluation attacks in the benchmark literature [2,3].

4.2 Evaluation Metrics

The primary evaluation metric is the Equal Error Rate (EER), computed at the detection threshold where FAR equals FRR. The secondary metric is min t-DCF, which accounts for the cost of countermeasure errors within a full ASV pipeline [3]. Both metrics are lower-is-better, and all results are reported on the LA evaluation partition rather than the development set.

4.3 Implementation Details

All models are implemented in PyTorch with torchaudio for audio loading. RawBoost augmentation is applied on-the-fly during training as a waveform-level transform. Official code repositories for RawNet2 and AASIST serve as reference implementations. The ASVspoof 2019 LA protocol files and .flac audio files are verified to be complete and consistent with the published dataset statistics prior to model training.

5. Preliminary Results and Current Progress

5.1 Completed Work

The following milestones have been achieved as of the midterm submission date:

- **Dataset acquisition and verification.** The full ASVspoof 2019 LA dataset has been downloaded, organized, and verified. Protocol files for all three splits have been parsed and confirmed complete. All audio files referenced in the protocol are present and accounted for.
- **Exploratory data analysis.** A detailed EDA notebook documents all dataset properties relevant to model design. The 89.8% class imbalance has been confirmed and will be addressed through weighted cross-entropy. The speaker-disjoint property across all three partitions has been verified programmatically (zero pairwise speaker overlap).
- **Architecture review and implementation planning.** All 18 core reference papers have been read, summarized, and integrated. Architecture components for RawNet2 and AASIST have been documented with reference to official implementations. Training hyperparameters have been fixed following the original papers.

5.2 Published Baselines

Model training is currently in progress. The table below reports published baseline numbers against which our reproductions will be compared. Our results will be added as training completes.

Model	Year	EER (%)	min t-DCF	Notes
CQCC + GMM [3]	2019	9.57	0.2366	Official challenge baseline
LFCC + GMM [3]	2019	8.09	0.2116	Official challenge baseline
RawNet2 [4]	2021	4.66	0.1294	Single system (fixed sinc)
AASIST [5]	2022	0.83	0.0275	Single system
AASIST-L [5]	2022	0.99	0.0309	Lightweight (85K params)
Resolution-Aware [15]	2026	0.16	—	Current SOTA
Our RawNet2 (reprod.)	2026	—	—	In progress
Our AASIST + SSL	2026	—	—	Planned

Table 2: Published baselines vs. our planned reproductions on ASVspoof 2019 LA evaluation set (lower is better).

5.3 EDA Findings Relevant to Modeling

- **Class imbalance is severe and consistent.** With 89.8% spoof samples across all splits, unweighted cross-entropy would produce a biased classifier. Weighted CE or OC-Softmax will be used.
- **Speaker-disjoint design confirmed.** Zero speaker overlap across train, development, and evaluation partitions ensures that speaker identity cannot be exploited as a detection shortcut.
- **Known-unknown attack split.** Training attacks A01–A06 include two voice conversion and four TTS systems. The evaluation introduces 13 entirely new attack types — the generalization stress test is built into the

benchmark protocol.

- **Dataset is complete and verified.** All audio files referenced in the protocol files are present and accounted for.

6. Future Work and Remaining Plan

6.1 Immediate Next Steps (Next Two Weeks)

The immediate priority is completing the RawNet2 baseline reproduction. Training will proceed using the hyperparameters in Section 3.5, with results validated against the published 4.66% EER single-system result [4]. Upon confirming reproducibility within ±0.5% EER, AASIST training will begin using the official implementation as a reference.

Concurrently, the SSL feature extraction pipeline will be implemented. wav2vec 2.0 features will be extracted from intermediate transformer layers following the layer-wise analysis in [13], with the extraction depth determined by a validation sweep over the development set. Fine-tuning will be performed end-to-end following the protocol of [8], which yielded a 10–20× improvement over frozen representations. RawBoost augmentation will be integrated and ablated: training runs with no augmentation, mode ① only, mode ② only, and combined ①+② will be compared.

6.2 Planned Experiments

- **Feature × Architecture grid:** All combinations of (raw sinc, SSL extracted, SSL fine-tuned) × (RawNet2, AASIST) — yielding six baseline configurations.
- **Augmentation ablation:** Each RawBoost mode evaluated independently and in combination with the strongest architecture.
- **Per-attack EER breakdown:** Individual EER reported for each of the 13 evaluation attacks (A07–A19) to identify which attack types pose the greatest challenge.
- **Cross-dataset evaluation:** The strongest trained model evaluated on the In-the-Wild dataset to measure the generalization gap following the protocol of [10].
- **Noise robustness evaluation:** Detection performance measured under controlled additive noise at SNR levels of 0, 5, 10, and 20 dB, motivated by [14].

6.3 Final Report Targets

The final report will present a complete ablation table across all feature, architecture, and augmentation combinations, accompanied by per-attack analysis and cross-dataset evaluation results. The project repository will be submitted with fully reproducible training code, pre-trained model weights, and evaluation scripts. Comparison against the resolution-aware 2026 SOTA [15] will contextualize the final system's performance relative to the current frontier. Emerging paradigms — including artifact-focused self-synthesis [18] and large audio-language model reasoning [17] — will be discussed as directions for future investigation beyond this project scope.

7. Conclusion

This work addresses the critical challenge of detecting synthetically generated speech in the face of increasingly sophisticated voice cloning technologies, including LLM-based TTS systems now evaluated against existing detectors [16]. Using the ASVspoof 2019 Logical Access benchmark [2], the project implements a structured comparison of end-to-end and SSL-augmented detection architectures, with RawBoost augmentation [9] targeting the generalization gap documented by Müller et al. [10]. As of the midterm, the dataset has been fully acquired and verified through comprehensive exploratory analysis, architecture implementations are underway, and the experimental plan is fully specified. The completed system will produce a systematic ablation study identifying which combinations of features, architectures, and augmentation strategies are most critical to cross-attack generalization — the central open challenge in deepfake audio detection research.

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